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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024 (Old)

Time: 03 Hours

Code: 100105 ***Subject: Workshop Manufacturing Practices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven only): **[14]**

(a) Which of the following is not an advantage of casting as a manufacturing process?
(i) Ability to create complex shapes
(ii) Low material waste
(iii) High dimensional accuracy
(iv) Suitable for mass production

(b) An Advanced Manufacturing Methods, Additive Manufacturing primarily differs from conventional subtractive methods in that:
(i) It uses more material
(ii) It removes material layer by layer
(iii) It builds components by adding material layer by layer
(iv) It only works with metals

(c) Which of the following is an example of an advanced manufacturing process?
(i) Forging
(ii) Turning
(iii) 3D Printing
(iv) Brazing

(d) Which of the following processes is considered a non-traditional machining process?
(i) Milling
(ii) Turning
(iii) Electro Discharge Machining (EDM)
(iv) Drilling

(e) Ultrasonic machining is best suited for
(i) Non-ferrous metal
(ii) Amorphous solids
(iii) Brittle materials
(iv) Dielectrics

(f) What is the primary purpose of a try square in carpentry?
(i) Measuring wood thickness
(ii) Sanding wooden surfaces
(iii) Checking right angles and marking straight lines
(iv) Driving nails into wood

(g) Which type of wood joint is best suited for making strong right-angle frame connections?
(i) Butt joint
(ii) Lap joint
(iii) Dovetail joint
(iv) Scarf joint

(h) Which of the following plastic moulding processes involves injecting molten plastic into a mould cavity?
(i) Blow moulding
(ii) Injection moulding
(iii) Compression moulding
(iv) Extrusion moulding

(i) Thermosetting plastics are preferred in moulding because:
(i) They can be remelted and reshaped
(ii) They retain their shape permanently after setting
(iii) They are biodegradable
(iv) They are flexible and soft

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(j) Before cutting glass, the surface should be:

(i) Painted
(ii) Coated with oil
(iii) Roughened with sandpaper
(iv) Marked and cleaned thoroughly

Q.2 Solve both questions : **[14]**

(a) Explain the comparison between Casting and Machining in terms of process, cost, accuracy, and material wastage.

(b) Explain Friction Stir Welding (FSW) as an advanced joining process. Describe its working principle.

Q.3 Solve both questions : **[14]**

(a) Explain the comparison between casting and forming with respect to their material applications, dimensional accuracy, and cost-effectiveness.

(b) Explain the comparison between Additive Manufacturing (3D Printing) and Subtractive Manufacturing in terms of material usage, design flexibility, and typical applications.

Q.4 Solve both questions : **[14]**

(a) Explain the step-by-step procedure to make a mortise and tenon joint.

(b) Discuss the importance of proper seasoning of wood in carpentry. What are the common methods used for seasoning wood?

Q.5 Solve both questions : **[14]**

(a) Describe the process of injection moulding. Include the materials used, steps involved, and key advantages.

(b) Compare thermoplastics and thermosetting plastics in the context of moulding. Provide examples of each.

Q.6 Solve both questions : **[14]**

(a) Explain the safety precautions and tools required for cutting glass accurately.

(b) Describe the step-by-step method to cut a straight line on a sheet of glass.

Q.7 Solve both questions : **[14]**

(a) Differentiate between arc welding and gas welding in terms of heat source, equipment, and applications.

(b) Describe the safety precautions to be followed during arc welding.

Q.8 Solve both questions : **[14]**

(a) Explain the brazing process and mention the common materials used for filler metal.

(b) List the advantages and limitations of brazing compared to welding.

Q.9 Write short notes on any two of the following: **[14]**

(a) CNC Machining

(b) Plastic Mouldings

(c) Soldering process

(d) Carpentry

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